

Project 1

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Project 1 – Data Cleaning & Preparation

Objective

The objective of this project is to clean and prepare a raw dataset by handling missing values, checking duplicate records, and correcting data formatting.

Tasks Performed

1. Missing Values Handling

- **Identified missing values in the CouponCode column.**
- **Replaced empty cells with No Coupon.**

2. Duplicate Checking

- **Checked duplicate records using OrderID.**
- **No duplicate values were found in the dataset.**

3. Data Formatting

- **Standardized the date format using: yyyy-mm-dd**
 - **Formatted numeric columns (UnitPrice and TotalPrice) to 2 decimal places.**
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Tools Used

- **Microsoft Excel**
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Final Result

- **Dataset cleaned successfully**

- No duplicate Order IDs
- Consistent date formatting
- Dataset ready for analysis

Before Cleaning

The dataset contained missing values in the CouponCode column and required formatting verification before analysis.

Issues Found:

- Missing values in CouponCode
- Date formatting needed standardization
- Numeric formatting required adjustment

OrderID	Date	Customer	Product	Quantity	Unit Price	Shipping Addment	Order Status	Marketing Num	Marketing Channel	Coupon Code	Referral Source	Total Price	
ORD0200	2023-01-04	C72649	Monitor	5	570.62	928	Main Debit Cai	Shipped	TRK3794	7	SAVEU	Instagram	2853.1
ORD0200	2024-08-23	C75739	Phone	2	151.25	823	Main Online	Shipped	TRK3918	3	SAVEU	Referral	302.7
ORD0200	2024-02-27	C81728	Tablet	5	550.68	512	Main Credit C	Canceled	TRK4298	8	FREESH	Email	2753.4
ORD0200	2023-10-15	C33540	Chair	1	273.19	275	Main Debit Cai	Returned	TRK6276	5	SAVEU	Faceboo	273.19
ORD0200	2025-05-08	C81840	Printer	4	628.01	688	Main Online	Deliverec	TRK2324	8	SAVEU	Email	2504.04
ORD0200	2023-10-23	C37249	Phone	2	245.86	934	Main Credit C	Shipped	TRK7233	4	SAVEU	Instagram	491.72
ORD0200	2025-06-17	C83492	Laptop	1	684.42	986	Main Gift Card	Returned	TRK3641	6	SAVEU	Faceboo	684.42
ORD0200	2023-05-12	C41460	Monitor	5	143.55	706	Main Cash	Shipped	TRK7880	9	FREESH	Faceboo	747.75
ORD0200	2025-04-02	C26817	Phone	2	134.28	804	Main Gift Card	Canceled	TRK6104	2	Email		268.56
ORD0200	2023-11-21	C31946	Desk	4	509.38	802	Main Credit C	Shipped	TRK3347	6	SAVEU	Google	2037.52
ORD0200	2023-12-28	C43443	Tablet	5	625.97	533	Main Credit C	Returned	TRK3885	9	WINTER	Instagram	3128.85
ORD0200	2024-02-15	C33861	Monitor	3	49.14	831	Main Online	Returned	TRK4822	7	SAVEU	Email	147.42
ORD0200	2024-10-15	C38785	Monitor	2	180.5	179	Main Debit Cai	Pending	TRK2041	6	FREESH	Referral	361
ORD0200	2023-08-30	C88349	Laptop	3	201.49	903	Main Credit C	Returned	TRK9191	4	FREESH	Faceboo	604.47
ORD0200	2023-02-27	C38474	Tablet	2	353.23	989	Main Debit Cai	Pending	TRK7395	3	SAVEU	Instagram	706.46
ORD0200	2023-07-17	C39416	Printer	1	473.96	942	Main Cash	Deliverec	TRK5493	3	Google		473.96
ORD0200	2023-01-24	C28976	Printer	2	533.81	914	Main Credit C	Pending	TRK13194	3	FREESH	Email	1067.62
ORD0200	2024-03-02	C80672	Tablet	1	423.4	382	Main Debit Cai	Shipped	TRK1937	3	Referral		423.4
ORD0200	2025-03-28	C94329	Desk	1	431.44	891	Main Gift Card	Pending	TRK3518	3	SAVEU	Faceboo	431.44
ORD0200	2023-01-23	C14952	Monitor	5	224	273	Main Cash	Returned	TRK5590	9	SAVEU	Google	1120
ORD0200	2023-05-08	C54829	Chair	5	62.51	587	Main Cash	Pending	TRK16718	9	WINTER	Instagram	312.55
ORD0200	2024-11-17	C39023	Monitor	4	342.95	569	Main Cash	Deliverec	TRK5264	6	Google		1371.8
ORD0200	2023-01-29	C24416	Monitor	2	49.62	533	Main Online	Pending	TRK7933	2	SAVEU	Google	99.24
ORD0200	2025-04-08	C42372	Phone	4	273.25	755	Main Debit Cai	Canceled	TRK3616	5	FREESH	Referral	1093
ORD0200	2024-10-30	C17470	Laptop	2	230.95	956	Main Online	Deliverec	TRK2877	3	WINTER	Referral	461.9
ORD0200	2024-07-24	C42577	Monitor	1	393.98	709	Main Debit Cai	Deliverec	TRK6394	3	WINTER	Email	393.98
ORD0200	2024-03-09	C21944	Laptop	5	59.55	286	Main Online	Deliverec	TRK3023	6	Referral		297.75
ORD0200	2024-02-22	C97593	Printer	1	472.74	881	Main Online	Shipped	TRK8911	6	WINTER	Google	472.74
ORD0200	2023-11-21	C59823	Phone	4	160.07	568	Main Online	Pending	TRK6585	5	Email		640.28
ORD0200	2024-09-28	C33260	Desk	3	423.53	830	Main Credit C	Deliverec	TRK5974	6	SAVEU	Referral	1270.59
ORD0200	2025-03-22	C42687	Monitor	2	428.7	895	Main Online	Pending	TRK5478	5	SAVEU	Referral	853.4
ORD0200	2023-06-30	C26445	Chair	3	500.08	996	Main Debit Cai	Returned	TRK3122	6	Faceboo		1500.24
ORD0200	2023-05-01	C12388	Tablet	5	536.72	830	Main Cash	Deliverec	TRK9926	10	Email		2683.6
ORD0200	2023-06-05	C52184	Chair	1	162.77	912	Main Online	Canceled	TRK7331	4	WINTER	Faceboo	162.77
ORD0200	2023-01-07	C67195	Chair	4	576.52	526	Main Credit C	Pending	TRK3584	5	WINTER	Faceboo	2306.08
ORD0200	2024-12-07	C93004	Monitor	2	465.15	129	Main Online	Shipped	TRK4688	2	SAVEU	Faceboo	930.3
ORD0200	2025-05-08	C59337	Laptop	2	534.43	465	Main Debit Cai	Pending	TRK5098	5	Faceboo		1068.86
ORD0200	2024-10-14	C10054	Laptop	2	76.74	409	Main Cash	Returned	TRK3400	7	SAVEU	Instagram	153.48
ORD0200	2024-12-18	C39324	Laptop	1	245.89	549	Main Debit Cai	Deliverec	TRK3482	3	SAVEU	Google	245.89
ORD0200	2025-02-27	C86954	Printer	3	256.40	968	Main Credit C	Deliverec	TRK3950	4	SAVEU	Referral	769.28
ORD0200	2023-08-28	C84766	Desk	2	344.02	267	Main Online	Deliverec	TRK7896	4	FREESH	Instagram	688.04
ORD0200	2023-12-05	C20212	Monitor	3	611.45	657	Main Gift Card	Pending	TRK6872	4	FREESH	Instagram	1834.35
ORD0200	2024-05-08	C28772	Printer	2	151.68	880	Main Cash	Canceled	TRK4077	6	Instagram		303.32
ORD0200	2023-12-18	C49572	Chair	3	498.21	971	Main Online	Canceled	TRK2035	8	Instagram		1497.63
ORD0200	2025-04-21	C73225	Phone	2	550.18	996	Main Gift Card	Returned	TRK7384	3	FREESH	Referral	1100.36
ORD0200	2024-02-10	C17388	Chair	1	317.25	157	Main Debit Cai	Returned	TRK4703	1	SAVEU	Faceboo	317.25
ORD0200	2024-06-26	C43232	Monitor	4	550.14	223	Main Debit Cai	Pending	TRK2510	4	FREESH	Instagram	2200.56
ORD0200	2023-07-15	C39154	Desk	2	209.94	323	Main Credit C	Returned	TRK4242	3	FREESH	Email	419.88
ORD0200	2023-08-26	C12408	Monitor	2	80.77	464	Main Debit Cai	Shipped	TRK3893	4	Google		161.54
ORD0200	2023-10-07	C11985	Printer	1	30.44	633	Main Credit C	Shipped	TRK5323	1	SAVEU	Email	30.44
ORD0200	2023-07-31	C36494	Tablet	1	179.22	446	Main Online	Canceled	TRK1467	6	FREESH	Instagram	179.22
ORD0200	2024-12-30	C36778	Tablet	1	389.33	434	Main Cash	Returned	TRK3761	2	SAVEU	Faceboo	389.33
ORD0200	2024-09-08	C30513	Tablet	1	586.44	232	Main Online	Returned	TRK6432	5	SAVEU	Faceboo	586.44
ORD0200	2024-05-18	C67951	Phone	4	488.78	512	Main Online	Returned	TRK3485	5	FREESH	Google	1955.12
ORD0200	2025-03-27	C48120	Chair	1	280.25	823	Main Online	Pending	TRK1019	5	WINTER	Email	280.25
ORD0200	2025-02-24	C35994	Printer	3	344.4	985	Main Online	Canceled	TRK3743	3	Faceboo		1033.2
ORD0200	2023-05-21	C54238	Tablet	3	475.21	566	Main Cash	Canceled	TRK6290	5	WINTER	Referral	1437.63
ORD0200	2023-01-22	C88036	Printer	5	314.27	842	Main Cash	Returned	TRK5247	6	SAVEU	Instagram	1571.35
ORD0200	2023-01-15	C60190	Phone	4	474.69	202	Main Gift Card	Canceled	TRK7088	6	SAVEU	Google	1898.76
ORD0200	2024-01-01	C37603	Chair	4	553.27	811	Main Online	Shipped	TRK4708	5	Faceboo		2233.08
ORD0200	2024-08-21	C33256	Desk	1	417.1	634	Main Debit Cai	Returned	TRK2341	3	FREESH	Referral	417.1
ORD0200	2024-03-24	C61618	Desk	4	122.2	104	Main Cash	Deliverec	TRK4348	7	FREESH	Faceboo	488.8
ORD0200	2023-08-22	C36497	Desk	2	43.15	691	Main Debit Cai	Pending	TRK1210	3	WINTER	Email	86.3
ORD0200	2025-02-02	C96895	Chair	1	564.86	913	Main Debit Cai	Shipped	TRK3422	6	WINTER	Referral	564.86
ORD0200	2025-02-02	C49320	Printer	3	130.61	782	Main Debit Cai	Canceled	TRK3957	4	FREESH	Faceboo	391.83
ORD0200	2023-12-30	C97735	Monitor	5	81.7	385	Main Credit C	Deliverec	TRK1174	6	Instagram		408.5
ORD0200	2024-11-15	C81804	Chair	5	127.18	715	Main Cash	Canceled	TRK1548	9	SAVEU	Referral	635.9

Chart1

Sheet1

Cleaned Dataset.xlsx

After Cleaning

The dataset was cleaned successfully using Microsoft Excel.

Improvements Made:

- Missing values replaced with No Coupon
- Duplicate records checked successfully
- Date format standardized to yyyy-mm-dd
- Numeric columns formatted to 2 decimal places

Final Status:

- No duplicate IDs
- Properly formatted dataset
- Ready for analysis

J	D	E	F	G	H	I	J	K	L	M	N
1	Product	Quantity	UnitPrice	ShippingAddress	PaymentMethod	OrderStatus	TrackingNumber	ItemsinCart	CouponCode	ReferralSource	TotalPrice
2	Monitor	5	570.62	928 Main St	Debit Card	Shipped	TRK37947903	7	SAVE10	Instagram	2853.10
3	Phone	2	151.35	823 Main St	Online	Shipped	TRK91186779	3	SAVE10	Referral	302.70
4	Tablet	5	550.68	512 Main St	Credit Card	Cancelled	TRK42903982	8	FREESHIP	Email	2753.40
5	Chair	1	273.19	275 Main St	Debit Card	Returned	TRK62788070	5	SAVE10	Facebook	273.19
6	Printer	4	626.01	668 Main St	Online	Delivered	TRK29241424	8	SAVE10	Email	2504.04
7	Phone	2	245.86	934 Main St	Credit Card	Shipped	TRK72976927	4	SAVE10	Instagram	491.72
8	Laptop	1	664.42	986 Main St	Gift Card	Returned	TRK96417362	6	SAVE10	Facebook	664.42
9	Monitor	5	149.55	706 Main St	Cash	Shipped	TRK78809193	9	FREESHIP	Facebook	747.75
10	Phone	2	134.28	904 Main St	Gift Card	Cancelled	TRK61042692	2	No Coupon	Email	268.56
11	Desk	4	509.38	102 Main St	Credit Card	Shipped	TRK33478363	6	SAVE10	Google	2037.52
12	Tablet	5	625.97	333 Main St	Credit Card	Returned	TRK98859248	9	WINTER15	Instagram	3129.85
13	Monitor	3	49.14	831 Main St	Online	Returned	TRK48234646	7	SAVE10	Email	147.42
14	Monitor	2	180.50	179 Main St	Debit Card	Pending	TRK20419991	6	FREESHIP	Referral	361.00
15	Laptop	3	201.49	903 Main St	Credit Card	Returned	TRK81913874	4	FREESHIP	Facebook	604.47
16	Tablet	2	393.33	980 Main St	Debit Card	Pending	TRK79186539	3	SAVE10	Instagram	786.66
17	Printer	1	473.96	942 Main St	Cash	Delivered	TRK54950938	3	No Coupon	Google	473.96
18	Printer	2	533.81	914 Main St	Credit Card	Pending	TRK13194259	3	FREESHIP	Email	1067.62
19	Tablet	1	423.40	382 Main St	Debit Card	Shipped	TRK11972116	3	No Coupon	Referral	423.40
20	Desk	1	431.44	891 Main St	Gift Card	Pending	TRK65191045	3	SAVE10	Facebook	431.44
21	Monitor	5	224.00	273 Main St	Cash	Returned	TRK59101365	9	SAVE10	Google	1120.00
22	Chair	5	62.51	587 Main St	Cash	Pending	TRK16718651	9	WINTER15	Instagram	312.55
23	Monitor	4	342.95	569 Main St	Cash	Delivered	TRK52646983	6	No Coupon	Google	1371.80
24	Monitor	2	49.62	633 Main St	Online	Pending	TRK75937350	2	SAVE10	Google	99.24
25	Phone	4	273.25	755 Main St	Debit Card	Cancelled	TRK36168083	5	FREESHIP	Referral	1093.00
26	Laptop	2	230.95	956 Main St	Online	Delivered	TRK28775057	3	WINTER15	Referral	461.90
27	Monitor	1	399.98	709 Main St	Debit Card	Delivered	TRK69423327	3	WINTER15	Email	399.98
28	Laptop	5	59.55	296 Main St	Online	Delivered	TRK80276812	6	No Coupon	Referral	297.75
29	Printer	1	472.74	881 Main St	Online	Shipped	TRK89117573	6	WINTER15	Google	472.74
30	Phone	4	160.07	568 Main St	Online	Pending	TRK65895539	5	No Coupon	Email	640.28
31	Desk	3	423.53	930 Main St	Credit Card	Delivered	TRK59744090	6	SAVE10	Referral	1270.59
32	Monitor	2	426.70	895 Main St	Online	Pending	TRK54751444	5	SAVE10	Referral	853.40
33	Chair	3	500.08	996 Main St	Debit Card	Returned	TRK31229981	6	No Coupon	Facebook	1500.24
34	Tablet	5	536.72	830 Main St	Cash	Delivered	TRK99261395	10	No Coupon	Email	2683.60
35	Chair	1	162.77	912 Main St	Online	Cancelled	TRK73312471	4	WINTER15	Facebook	162.77
36	Chair	4	576.52	926 Main St	Credit Card	Pending	TRK55467915	5	WINTER15	Facebook	2306.08
37	Monitor	2	465.15	128 Main St	Online	Shipped	TRK46850888	2	SAVE10	Facebook	930.30
38	Laptop	2	534.43	465 Main St	Debit Card	Pending	TRK50961250	5	No Coupon	Facebook	1068.86
39	Laptop	2	76.74	409 Main St	Cash	Returned	TRK94005261	7	SAVE10	Instagram	153.48
40	Laptop	1	245.89	549 Main St	Debit Card	Delivered	TRK14895975	3	SAVE10	Google	245.89
41	Printer	3	256.46	666 Main St	Credit Card	Delivered	TRK85508960	4	SAVE10	Referral	769.38
42	Desk	2	344.02	267 Main St	Online	Delivered	TRK78967923	4	FREESHIP	Instagram	688.04
43	Monitor	3	611.45	657 Main St	Gift Card	Pending	TRK69753327	4	FREESHIP	Instagram	1834.35
44	Printer	2	151.66	980 Main St	Cash	Cancelled	TRK40772558	6	No Coupon	Instagram	303.32
45	Chair	3	499.21	871 Main St	Online	Cancelled	TRK20351666	8	No Coupon	Instagram	1497.63

Conclusion

This project helped me understand the basics of data cleaning and preparation using Excel. I learned how to handle missing values, verify duplicate data, and standardize dataset formatting before analysis.